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Project Title:	Advanced Life Support: Development of the M2BR for Biological Treatment of Wastewater Generated During Long-Term Space Missions		
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Contact Monitor:		**Contact Phone:**	
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	Advanced life support (ALS) provides basic life-sustaining functions inside the spacecraft or on planetary surfaces, and includes providing usable water and breathable air, and managing wastes. This research proposal focuses on the development of technology to recover potable water while sustaining the appropriate levels of habitability. Wastewater is the most voluminous waste stream in human spacecrafts and thus there is a substantial need to conserve and/or recycle this water on long-term space flights. Urine, atmospheric condensate, and spent hygiene water contain substantial concentrations of organic and nitrogenous pollutants. Conventional biological wastewater treatment processes (i.e., those used on Earth) adequately remove carbonaceous pollutants, but often suffer from poor nitrogenous pollutant removal. There is therefore a need to develop high quality wastewater treatment systems that remove both carbonaceous and nitrogenous pollutants for the success of long-term space flights. We propose the application of a novel membrane bioreactor for the treatment of wastewaters generated on space flights. The proposed treatment technology will use a		

Task Description:	membrane to transfer oxygen to microorganisms that will remove the carbonaceous and nitrogenous pollutants in the wastewater and a second membrane to achieve completely cell-free treatment effluent. The proposed membrane-coupled, membrane-aerated (M2BR) design should retain the attractive features of the conventional membrane-coupled bioreactor while additionally offering highly efficient nitrogenous pollutant removal, which is necessary for long-term space missions. The proposed M2BR will be particularly advantageous because it will have excellent treatment performance in a single bioreactor. Our proposed application of the M2BR is substantially different from conventionally operated membrane-coupled bioreactors. The membrane aeration device will be designed to ensure that oxygen only partially penetrates the biofilm, such that the outer portion of the biofilm and bulk fluid will be anoxic. This unique operating regime will lead to simultaneous nitrification-denitrification. The use of oxygen by the biofilm will be primarily for nitrification while the majority of carbonaceous pollutant removal will be due to anaerobic metabolic processes. The goal of the proposed research is to develop that the M2BR can effectively treat a synthetic space flight wastewater at the bench-scale.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project further developed two technologies that are used on terrestrial Earth. The first is the membrane-coupled biroeactor (MBR), which utilizes a membrane filter to retain high biomass densities in an otherwise conventional bioreactor. The second is the membrane-aerated bioreactor (MABR), which utilizes a microporous membrane to oxygenate wastewater treatment bioreactors. Both of these technologies are currently under development in the field of biological wastewater treatment; our research has helped further this development. Furthermore, the combination of these two technologies to become the membrane-coupled, membrane-aerated bioreactor (M2BR) could also have terrestrial applications, such as on Navy ships.
Task Progress:	Potable water is a critical resource for human beings during long-term space travel. While spaceships are currently fueled by the combustion of H2 and O2, which generates sufficient quantities of potable water, future long-term space missions (e.g., to Mars) will utilize nuclear-based power, thus eliminating this source of water. Future long-term space missions, therefore, will need to treat and reuse as much water and wastewater as possible. Wastewater treatment during long-term spaces missions poses unique challenges compared to analogous terrestrial systems. Wastewater treatment must produce a very high quality effluent without adversely affecting the habitability of the spacecraft. Another challenge is to design the entire process to be compatible with microgravity conditions. Conventional aerobic biological wastewater treatment, for example, depends on gravity to create buoyant air bubbles for aeration and on gravitational clarifiers to separate the biomass from the treated effluent. In this study, we tested the ability of the M2BR to treat a synthetic wastewater formulated by NASA to simulate the combination of urine, atmospheric condensate, and used hygiene water that would be generated during space missions. The NASA wastewater ersatz contains relatively high concentrations of both nitrogenous (700 mg N l-1) and carbonaceous pollutants (chemical oxygen demand = 800 mg l-1). Our goals for this study were to identify an appropriate start-up procedure so that the M2BR could achieve simultaneous nitrification/denitrification and to determine the limitations of the M2BR in treating this wastewater. The first M2BR experiment was fed the full-strength NASA wastewater ersatz for 66 days at a hydraulic residence time (HRT) of 24 hours without biosolids removal, simulating the simplest start-up procedure for the M2BR. The effluent COD concentration steadily declined during the first 30 days of operation, after which the M2BR performance was reasonably consistent (mean COD removal efficiency = 87%). In contrast, no ammonia oxidation was observed throughout the entire experiment. Suspended biomass accumulated rapidly during the first 25 days, after which it gradually increased until the end of the experiment. Because the first M2BR experiment failed to achieve nitrification, the second M2BR experiment involved a more complicated start-up procedure to ensure better performance. The second M2BR experiment, therefore, involved four distinct phases of operation during which the composition of the feed solution was varied while the HRT was maintained at 24 hours. The M2BR was initially fed an ammonia-only solution (Phase 1: 29 days; COD = 0 mg l-1; NH3 = 200 mg l-1 as N) to establish the nitrifying bacterial community, after which a dilute version NASA wastewater ersatz was fed to the M2BR to also establish the heterotrophic bacterial community (Phase 2: 23 days; COD = 220 mg l-1; NH3 = 170 mg l-1 ¬as N). After high removals of both COD and NH3 were observed, then the feed solution was changed to a more concentrated version of the NASA wastewater ersatz that had a COD:N ratio that would support simultaneous nitrification/denitrification in the M2BR (Phase 3: 33 days; COD = 480 mg l-1; NH3 = 180 mg l-1 as N). During the final phase of the second M2BR experiment, the full-strength NASA wastewater ersatz was fed to the bioreactor (Phase 4: 20 days; COD = 780 mg l-1; NH3 = 700 mg l-1 ¬as N). During Phase 1, ammonia removal efficiency was initially low but eventually was sustained at > 90%; an apparent stoichiometric conversion of the ammonia to nitrate was observed during this phase. During Phase 2, COD removal was consistently excellent while the ammonia removal efficiency was sustained. The concentration of nitrate in the M2BR effluent decreased by about 35% during Phase 2 compared to Phase 1, indicating that denitrification was occurring to a minor extent. During Phase 3, COD and NH3 removal remained excellent, but a substantial reduction in the nitrate levels was observed in the effluent of the M2BR – corresponding to better than 90% removal of both COD and total nitrogenous species. During Phase 4, ammonia concentrations in the effluent increased substantially, although COD removal remained excellent. Although the second experiment demonstrated that the M2BR could achieve excellent COD and total nitrogenous pollutant removal during Phase 3, a substantial decline in nitrification/denitrification efficiency was observed once the feed solution was switched to the full-strength NASA wastewater ersatz during Phase 4. We assumed that this decline in M2BR performance occurred because the oxygen demand for treatment exceeded the OTR of the gas-transferring membrane. The third M2BR experiment, therefore, was performed to evaluate the performance of the M2BR using pure oxygen to increase the OTR to the bioreactor. Results were generally similar to the second M2BR experiment, although the additional oxygen transfer inhibited denitrification under certain conditions (insufficient electron donor). Our research demonstrates that the gas transferring membrane is a key design constraint that will control the performance of the M2BR. The challenge with membrane-aerated biofilms for simultaneous nitrification and denitrification is to ensure that there is sufficient gas transfer to promote complete ammonia oxidation, but not so much gas transfer that the biofilm is fully penetrated with oxygen, thus inhibiting denitrification. In our experiments, identifying the optimum rate of gas transfer proved difficult in that oxygen transfer during many of our experimental

	conditions was either excessive (e.g., Experiment 2, Phase 2) or deficient (e.g., Experiment 2, Phase 4). To resolve this issue, future research is needed to calibrate mathematical models of membrane-aerated biofilms with analogous experimental analyses. Our research also demonstrates that the membrane filter serves as a key hygienic constraint for astronauts that will need to live in close proximity to this biological treatment process throughout the duration of their long-term space mission. The relatively large quantity of suspended biomass that accumulated in the M2BR would have otherwise been discharged with the effluent. During space missions, this biomass would then have a substantially better opportunity to adversely affect astronaut health. It remains unclear, however, whether this suspended biomass helps catalyze pollutant removal or it is inactive biomass that is merely accumulating in the M2BR.
Bibliography Type:	Description: (Last Updated: 02/27/2008)
Abstracts for Journals and Proceedings	Chen R, Semmens MJ, LaPara TM. "The development of the membrane-aerated, membrane-coupled bioreactor (M2BR) for wastewater treatment in space." Habitation 2006 Conference, Orlando, FL, February 5-8, 2006. Habitation 2006;10(3-4):157. , Jan-2006
Articles in Peer-reviewed Journals	Chen RD, Semmens MJ, Lapara TM. "Biological treatment of a synthetic space mission wastewater using a membrane-aerated, membrane-coupled bioreactor (M2BR)." J Ind Microbiol Biotechnol. 2008 Jan 15; [Epub ahead of print] PMID: 18202862 , Jan-2008
Dissertations and Theses	Chen RD. "Membrane bioreactors for wastewater treatment." Dissertation, University of Minnesota, Twin Cities, October 2006. , Oct-2006